

AEGIS Financial Model — 3-Year Projection

Last updated: 2026-02-26

Assumptions

Market Assumptions

- Total addressable market (TAM): ~\$600M (health data de-identification, 2026 est.)
- Serviceable addressable market (SAM): ~\$120M (medical imaging de-identification + research data sharing, US + EU)
- Serviceable obtainable market (SOM): ~\$2M–5M Year 1, growing to ~\$10M–20M Year 3
- NIH DMS policy is the primary demand catalyst for research customers
- Enterprise radiology de-identification is the expansion market (Phase 5+)

Product Assumptions

- Open-source core available throughout (builds adoption, not directly monetized)
- Hosted SaaS is the primary revenue driver
- Free tier converts at 10–15% to paid within 6 months
- Average contract length: 12 months (annual billing default)
- No on-premises license revenue in Year 1; begins Year 2

Cost Assumptions

- Founding team: 2–3 people (Year 1), growing to 8–12 (Year 3)
- Cloud infrastructure costs scale at ~15–25% of SaaS revenue
- Cloud AI costs are passed through at 15–20% margin
- Customer acquisition cost (CAC): ~\$2,000–5,000 (low-touch for research, high-touch for enterprise)
- Gross margin target: 70–80% on SaaS, 40–60% on services

Revenue Model

Year 1 — Foundation (\$120K–350K ARR)

Focus: Academic labs, small multi-site studies, NIH-funded researchers. Land with free tier, convert to Research tier.

Revenue Stream	Customers	ARPA/mo	Monthly Rev	Annual Rev
Research tier (\$499/mo)	8–15	\$499	\$4K–7.5K	\$48K–90K
Consortium tier (\$1,999/mo)	2–4	\$1,999	\$4K–8K	\$48K–96K
Enterprise tier (custom)	0–1	\$5,000	\$0–5K	\$0–60K
Add-ons (DIMSE, support)	4–8	\$200	\$0.8K–1.6K	\$10K–19K
Professional services	—	—	\$0.5K–2K	\$6K–24K
Storage overage	—	—	\$0.3K–1K	\$4K–12K
Cloud AI pass-through	—	—	\$0.2K–0.5K	\$2K–6K

Total Year 1	10–20 customers			\$118K–\$307K

Milestone targets:

- Q1: Launch hosted SaaS, 50+ free tier signups, 3–5 paid conversions
- Q2: First consortium customer (multi-site study), 8–10 total paid
- Q3: 100+ free tier, 12–15 paid, first professional services engagement
- Q4: 150+ free tier, 15–20 paid, pipeline for 2–3 enterprise prospects

Year 2 — Growth (\$500K–1.5M ARR)

Focus: Expand consortium base, land first enterprise deals, begin on-premises licensing. Hire dedicated sales (enterprise) and customer success.

Revenue Stream	Customers	ARPA/mo	Monthly Rev	Annual Rev
Research tier	25–40	\$499	\$12.5K–20K	\$150K–240K
Consortium tier	8–15	\$1,999	\$16K–30K	\$192K–360K
Enterprise tier	3–6	\$7,500	\$22.5K–45K	\$270K–540K
On-premises licenses	1–2	—	—	\$25K–75K
Add-ons	15–25	\$250	\$3.8K–6.3K	\$45K–75K
Professional services	—	—	\$2K–5K	\$24K–60K
Storage overage	—	—	\$1.5K–4K	\$18K–48K
Cloud AI pass-through	—	—	\$1K–3K	\$12K–36K
Total Year 2	37–63 customers			\$736K–\$1,434K

Milestone targets:

- Q1: First enterprise customer signed (hospital network or pharma)
- Q2: 30+ paid customers, first on-premises license inquiry
- Q3: 200+ free tier, DIMSE adoption growing, first renewal cohort (measure NRR)
- Q4: 50+ paid, enterprise pipeline 5+ prospects, first international customer

Year 3 — Scale (\$2M–5M ARR)

Focus: Enterprise expansion, international markets (EU/UK/ANZ), product-led growth flywheel established. Team of 8–12.

Revenue Stream	Customers	ARPA/mo	Monthly Rev	Annual Rev
Research tier	50–80	\$499	\$25K–40K	\$300K–480K
Consortium tier	20–35	\$1,999	\$40K–70K	\$480K–840K

Enterprise tier	8–15	\$10,000	\$80K–150K	\$960K–1,800K
On-premises licenses	3–6	—	—	\$75K–300K
Add-ons	35–60	\$300	\$10.5K–18K	\$126K–216K
Professional services	—	—	\$5K–12K	\$60K–144K
Storage overage	—	—	\$5K–12K	\$60K–144K
Cloud AI pass-through	—	—	\$3K–8K	\$36K–96K
Protocol template packs	10–20	—	—	\$15K–30K
Total Year 3	81–140 customers			\$2,112K–\$4,050K

Cost Structure

Year 1

Category	Monthly	Annual	Notes
Team (2–3 people)	\$25K–40K	\$300K–480K	Founders + first hire (eng or sales)
Cloud infrastructure	\$1.5K–4K	\$18K–48K	GCP/AWS for hosted SaaS
Cloud AI costs (pass-through)	\$0.2K–0.5K	\$2K–6K	Gemini, Vision, Textract usage
SaaS tools	\$0.5K–1K	\$6K–12K	Stripe, monitoring, email (Brevo), domain
Legal / compliance	—	\$10K–20K	BAA templates, privacy policy, terms of service
Marketing	\$0.5K–1K	\$6K–12K	Conference travel, content, ads
Total Year 1 costs		\$342K–\$578K	

Year 2

Category	Monthly	Annual	Notes
Team (5–7 people)	\$60K–95K	\$720K–1,140K	+2–3 eng, +1 sales, +1 customer success
Cloud infrastructure	\$5K–15K	\$60K–180K	Scales with customer base
Cloud AI costs	\$1K–3K	\$12K–36K	

SaaS tools	\$1K–2K	\$12K–24K	
Legal / compliance	—	\$20K–40K	SOC 2 audit prep, BAA execution
Marketing	\$2K–5K	\$24K–60K	Conferences (RSNA, OHBM, SfN), content
Office / misc	\$1K–2K	\$12K–24K	
Total Year 2 costs		\$860K–\$1,504K	

Year 3

Category	Monthly	Annual	Notes
Team (8–12 people)	\$100K–170K	\$1,200K–2,040K	+2 eng, +1 sales, +1 support, +1 product
Cloud infrastructure	\$15K–40K	\$180K–480K	Multi-region, enterprise SLA
Cloud AI costs	\$3K–8K	\$36K–96K	
SaaS tools	\$2K–4K	\$24K–48K	
Legal / compliance	—	\$40K–80K	SOC 2 Type II audit, international DPAs
Marketing	\$5K–12K	\$60K–144K	Event sponsorships, demand gen
Office / misc	\$2K–4K	\$24K–48K	
Total Year 3 costs		\$1,564K–\$2,936K	

P&L Summary (3-Year)

	Year 1	Year 2	Year 3
Revenue	\$120K–\$307K	\$736K–\$1,434K	\$2,112K–\$4,050K
COGS (infra + AI)	\$20K–\$54K	\$72K–\$216K	\$216K–\$576K
Gross profit	\$100K–\$253K	\$664K–\$1,218K	\$1,896K–\$3,474K
Gross margin	82%	85%	86%
Operating expenses	\$322K–\$524K	\$788K–\$1,288K	\$1,348K–\$2,360K
Net income (loss)	(\$222K)–(\$271K)	(\$124K)–(\$70K)	\$548K–\$1,114K
Net margin	-88% to -185%	-5% to -17%	26% to 28%

Break-even: Mid to late Year 2 (monthly basis), full-year profitability in Year 3.

Unit Economics

Customer Lifetime Value (LTV)

Tier	ARPA/yr	Avg. lifetime (yr)	Gross margin	LTV
Research	\$5,988	2.5	82%	\$12,275
Consortium	\$23,988	3.0	85%	\$61,169
Enterprise	\$90,000	3.5	85%	\$267,750

Customer Acquisition Cost (CAC)

Tier	Channel	Est. CAC	LTV:CAC
Research	Product-led (free tier → conversion)	\$1,500–2,500	5–8x
Consortium	Outbound + conference + referral	\$5,000–10,000	6–12x
Enterprise	Direct sales + POC	\$15,000–30,000	9–18x

Target LTV:CAC ratio: >3x (healthy SaaS benchmark). All tiers exceed this.

CAC Payback Period

Tier	CAC	Monthly margin	Payback
Research	\$2,000	\$410	4.9 months
Consortium	\$7,500	\$1,699	4.4 months
Enterprise	\$22,500	\$6,375	3.5 months

All tiers achieve payback within 5 months — well under the 12-month SaaS benchmark.

Key Metrics Dashboard

Metric	Year 1 Target	Year 2 Target	Year 3 Target
ARR	\$120K–\$307K	\$736K–\$1,434K	\$2,112K–\$4,050K
Total customers (paid)	10–20	37–63	81–140
Free tier users	150+	300+	500+
Free → paid conversion	10%	13%	15%
Net revenue retention (NRR)	100%	110%	120%
Monthly churn	<5%	<4%	<3%

Gross margin	82%	85%	86%
LTV:CAC	5x+	8x+	10x+
CAC payback	<6 months	<5 months	<4 months
Headcount	2–3	5–7	8–12
Studies processed (platform total)	50K	300K	1M+

Funding Scenarios

Scenario A: Bootstrapped

- No external funding
- Team stays 2–3 through Year 1
- Revenue covers costs by mid-Year 2
- Slower growth but full ownership retained
- **Risk:** Can't hire sales early enough, miss enterprise window
- **Requires:** \$300K–500K initial capital (savings or angel)

Scenario B: Seed Round (\$1M–2M)

- Raise after first 5–10 paying customers (proof of product-market fit)
- Hire 2–3 engineers + 1 sales immediately
- Accelerate enterprise pipeline
- **Use of funds:** Engineering (50%), sales (25%), cloud infra (15%), legal/compliance (10%)
- **Expected dilution:** 15–25%
- **Investors:** Health-tech focused angels, SBIR/STTR grants, early-stage VCs (Rock Health, Launchpad Digital Health)

Scenario C: Series A (\$5M–10M, Year 2–3)

- Raise after \$500K+ ARR and 3+ enterprise customers
- Scale sales team (3–5 reps), expand engineering (10+), international expansion
- SOC 2 Type II certification
- **Expected dilution:** 20–30%
- **Investors:** Health-tech VCs (a16z Bio, General Catalyst, Bessemer)

Scenario D: Grant-Funded (Non-Dilutive)

- NIH SBIR/STTR Phase I (\$275K, 6 months) → Phase II (\$1M, 2 years)
- NSF SBIR (\$275K Phase I, \$1M Phase II)
- ARPA-H (Advanced Research Projects Agency for Health) — larger grants possible
- NIST cybersecurity/privacy grants
- **Advantage:** Non-dilutive, validates product with federal stamp of approval
- **Disadvantage:** Slow timeline (6–12 months from application to award), reporting overhead
- **Can be combined with Scenario A or B**

Sensitivity Analysis

Bull Case (Year 3: \$5M+ ARR)

- NIH DMS enforcement drives rapid adoption

- Land 2+ large pharma enterprise deals (\$15K+/mo each)
- International expansion (EU GDPR creates same demand)
- NRR > 130% (customers expand from Research → Consortium → Enterprise)

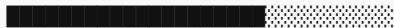
Base Case (Year 3: \$2.5M–3.5M ARR)

- Steady academic adoption, 2–4 enterprise customers
- 15% free-to-paid conversion
- NRR ~110–120%
- Break-even by late Year 2

Bear Case (Year 3: \$1M–1.5M ARR)

- Slower enterprise adoption (long sales cycles)
- Free tier cannibalizes paid (self-hosted open source is "good enough")
- Competitor (Flywheel or new entrant) adds defacing
- NRR < 100% (churn exceeds expansion)
- **Mitigation:** Reduce burn, focus on Research tier, pursue SBIR grants

Revenue Diversification Over Time

Year 1:		SaaS 75% Services 15% Overage 10%
Year 2:		SaaS 72% On-prem 5% Services 13% Overage 10%
Year 3:		SaaS 68% On-prem 9% Services 10% Overage 8% Templates 5%

SaaS subscriptions remain the dominant revenue stream throughout, with on-premises licensing becoming meaningful in Year 3 as enterprise and government customers require air-gapped deployments.

Implementation Priorities for Monetization

Immediate (before launch)

1. Stripe integration for subscription billing
2. Plan enforcement middleware (study count, storage, project limits)
3. Usage metering (study count per month, storage bytes per project — already tracked via existing APIs)
4. Pricing page on landing site

Near-term (Year 1)

5. Self-service plan upgrade/downgrade (Stripe Billing Portal)
6. Invoicing workflow for institutional POs (Net-30)
7. Usage dashboard for customers (current usage vs. plan limits)
8. Grant budget letter generator (PDF export)

Medium-term (Year 2)

9. SOC 2 Type II audit
10. Enterprise quoting and contract management
11. On-premises license packaging (Docker + Terraform bundle)
12. Multi-currency billing (EUR, GBP)